

Hydrophilic tyrosine-based phenolic resin with micro-ripples morphology for marine antifouling application

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ABSTRACT

Since biofouling challenges negatively influence the marine and transportation industries, developing effective antifouling materials have attracted extensive concern. A tyrosine-based antifouling phenolic resin (TPP resin) was synthesized using tyrosine as a natural phenol source. TPP exhibited shell-like surface morphology with micro-ripples and excellent anti-adhesion properties against bacteria and diatom. The micro-ripples surface might be caused by the strong hydrogen bonding or ionic interaction among tyrosine units resulting in micro-phase separation during the curing process. Tyrosine content in TPP resin has a great influence on the surface properties, morphology and antifouling characteristics. The higher the tyrosine content, the higher is the surface hydrophilicity, the denser and more regular is the micro-ripples morphology, and the stronger is the antifouling performance. TPP-60 % exhibited the best antifouling performance. Combination of the surface hydrophilicity and regular micro-ripples surface morphology afford TPP excellent antifouling performance. TPP resins offer a broad prospect for developing phenolic resin in the antifouling field.

1. Introduction

The adverse accumulation of microorganisms, algae, and animals on structures immersed in marine water is known as marine fouling [1,2]. When biofouling forms in the marine environment, several negative effects are generated, including accelerating the corrosion of marine equipment and ideally affecting its use, thereby causing severe economic losses [3] and directly and indirectly affecting normal human activities in the ocean [4,5]. Various materials have been employed to investigate effective marine antifouling strategies, such as engineered microtopography surfaces, super-hydrophilic surfaces, fouling-degrading, fouling-release coatings, etc. on the basis of the adhesion mechanism of marine organisms on the surface of materials. Researchers frequently draw inspiration for biomimetic materials from natural biological morphology to combat fouling [1,6–8]. Yanyan Zhang et al. replicated the boundary structures of taro leaf and sharkskin, and then measured antifouling characteristics by recording the settlement amount of fouling microalgae [9]. By investigating the effect of

boundary structures on the antifouling performance, they discovered that the shark skin replica with V-groove riblets and imbricate boundary structure exhibited the best antifouling performance [8–12]. Furthermore, hydrophilic materials have recently attracted increased attention because of their functionality with molecular design freedom and chemical diversity [13], nontoxic safety, and excellent antifouling property. The hydrophilic polymer generates a strong hydration barrier on the surface that effectively prevents nonspecific adsorption. Abdullah M. Alswieleh et al. prepared amino acid-derived zwitterionic hydrophilic polymer brushes, which were initiated from the surface of silicon wafers. The brushes exhibited stimulus-responsive behavior and negligible cytotoxicity and biofouling resistance [14]. While both materials have their advantages, efforts are put together to find alternative strategies to combine biomimetic structure and surface chemistry to prepare effective and environmental antifouling materials, which offer a promising direction for developing a new generation of antifouling materials [15,16].

Phenolic resin has a long development history with several desirable

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characteristics [17,18] and has been widely employed in numerous fields [19–21]. The traditional phenolic resin is synthesized by condensation of phenol and formaldehyde. However, phenol is toxic and carcinogenic, making it unsustainable as a product of the petroleum industry. Due to environmental protection and sustainability, it is critical to develop renewable natural resources and low-cost substitutes for phenol to synthesize new phenolic resin and give it unique characteristics to expand the application of phenolic resin [22,23]. Recently, natural renewable phenolic compounds have gradually replaced phenol to prepare eco-friendly and sustainable phenolic resins. Jiongjiong Li et al., for instance, fabricated tannin-based phenolic resins using depolymerized natural Acacia mangium tannin and discovered that these resins exhibited excellent adhesion performance and high thermal stability that have great potential for industrial applications [19].

Amino acids are sustainable raw materials that are cheap and readily available. Tyrosine is the only conditionally essential amino acid having a phenolic hydroxyl group, allowing it to be employed as a phenolic source to prepare phenolic resin. As a hydrophilic compound, tyrosine can be widely found in the daily food intake of human beings and is readily generated by various routines [22]. In addition, tyrosine plays an indispensable role in the metabolism, growth, and development of humans and animals. To the best of our knowledge, minimal research exists on the introduction of tyrosine in the preparation of phenolic resins for antifouling.

Tyrosine-based thermosetting phenolic resin (TPP resin) with excellent antifouling performance was synthesized using tyrosine as the natural phenol source in this study. The chemical structure, elemental analysis, thermostability, hydrophilicity, and surface morphology of TPP resins were investigated and evaluated. Furthermore, we employed a spread plate method, biofilm formation experiment, and laboratory static diatom anti-adhesion assay to investigate the antifouling effect of selected bacteria and diatom as model pollutants. The results indicated that the TPP resins exhibited efficient antifouling performance. Overall, the development of novel phenolic resins shows potential applications in marine antifouling and other relevant aspects.

2. Materials and methods

Materials, characterization methods, physical properties of the resins are available in the SI. Thermosetting TPP resins were prepared by polymerization and curing with tyrosine, phenol and paraformaldehyde as raw material, NaOH as catalyst for the reaction, details seen in the SI.

2.1. Bacterial adhesion and biofilm formation

The spread plate method was used for the number of adhered bacteria on the material surface as the quantitative analysis. The number of colonies on the agar plate reflected the antifouling performance of TPP resins [24,25]. A shaker was used to imitate a fluid environment, thus testing the short-time dynamic antifouling property of the resins [26]. Resins were incubated in bacterial suspension (20 mL) for 24 h at 37 °C under 100 rpm shaking. After that, the resins were gently cleaned with PBS twice to remove loosely adhered bacterial cells and placed in 10 mL PBS buffer solution, then ultrasonicated for 30 min. The PBS solution of bacteria extracted from the surface was diluted and spread onto an agar plate for overnight incubation at 37 °C. The images of bacterial colonies on the agar plates were observed and counted, and each sample was performed in triplicate.

We used SEM to observe the static bacterial biofilm formation on the resin surface. Resins were incubated in bacterial suspension for 7 days at 37 °C and then washed gently with PBS buffer solution. Resins that adhered to bacteria were immersed into 2.5 % glutaraldehyde for fixing a night at 4 °C in the dark condition, then cleaned with PBS buffer solution for cleaning off remaining glutaraldehyde, followed by dehydration successively with a series of graded ethanol solutions (10 %, 30 %, 50 %, 70 %, 90 %, and 100 %). After the sample was dried, the

surface was coated with gold for SEM observation.

2.2. Diatom settlement experiments

Nitzschia Closterium (*N.closterium*) was chosen to evaluate the antifouling property of the resins. *N.closterium* suspension was put into conical flasks with TPP resins statically settled for 2 days, 7 days and 14 days (12 h/12 h light-dark) at 22 °C, samples were washed with filter-sterilized seawater to remove unattached diatoms [5,9,27]. The adhesion of *N.closterium* on resins surface was observed by Fluorescence Microscope, which record three random fields (20 × magnification, 0.32 mm²/per field) of view for each sample. The quantity of *N. closterium* settlement results was counted using ImageJ software shown as numbers/mm². Data from at least three independent tests were expressed as the mean ± SD.

3. Results and discussions

Tyrosine, as a natural source of phenolic compound, was selected to partly replace phenol for TPP resins synthesis. The chemical structure, elemental analysis, thermostability, hydrophilicity and surface morphology of the resins were evaluated. The antifouling properties of the TPP resins were studied as follows.

3.1. Chemical structure analysis

Fig. 1(a) demonstrates the FT-IR spectra of the cured TPP resins. As is shown, the spectra of all the TPP resins are considerably similar to that of the neat phenolic resin because of their similar chemical structures. The stretching vibrations of Ph-OH, O-H of -COOH, and typical N-H bands of tyrosine are responsible for the wide peak range of 3300–3500 cm⁻¹. The formation of inter and intramolecular hydrogen bonds was attributed to the wide peak [28,29]. The absorption bands at 2922 cm⁻¹ and 2828 cm⁻¹ were related to the C-H stretching vibrations. Meanwhile, the peak intensities at 1219 cm⁻¹, 1470 cm⁻¹, and 596 cm⁻¹ were increased after adding tyrosine, which could be attributed to the stretching vibrations absorption of C-O, C-N, and C=O. The absorption peak of the C-O bond in the carboxyl group is 1150 cm⁻¹. The phenolic resins exhibited typical aromatic rings in the region of 1350–1600 cm⁻¹, and the peaks between 750 and 870 cm⁻¹ are because of the vibration of aromatic C-H. [19]. The results indicated that the TPP resins were successfully prepared and employed to provide further evidence for the antifouling performance.

XPS was used to investigate the surface chemical elements and composition characteristics of the resin. In Fig. 1(b), TPP resins have several C and O elements on the surface, with the peaks appearing at 283.9 and 531.8 eV. Table S2 shows that with the increase of tyrosine, the content of N elements also increased (0.54, 1.67, 1.34, and 2.97 for TPP-0 %, TPP-20 %, TPP-40 %, and TPP-60 %, respectively). In the O 1 s spectrum of TPP-0 %, the peaks at 533.3 eV correspond to C-OH in Fig. 1(c), while in O 1 s of TPP-60 % (seen in Fig. 1(d)) can mainly be separated into three peaks at 533.3, 532.5, and 531.4 eV due to phenolic hydroxyl C-OH, O=C-O, and O=C-O [30]. In addition, the surface element distribution of the TPP resins was characterized using EDS, as shown in Fig. S3. The elements N and O are the most obvious and widely distributed on the surface of TPP-60 %. These results confirmed that the curing process of resin did not break the chemical structure of tyrosine, and the TPP resins were synthesized.

3.2. Thermal property

The thermal stability of materials was a crucial parameter for assessing organic polymer performance. We analyzed the influence of tyrosine content on the thermal stability of the TPP resins using TG curves in Fig. 2. According to previous research, the thermal degradation of phenolic resin was divided into the following three stages: the

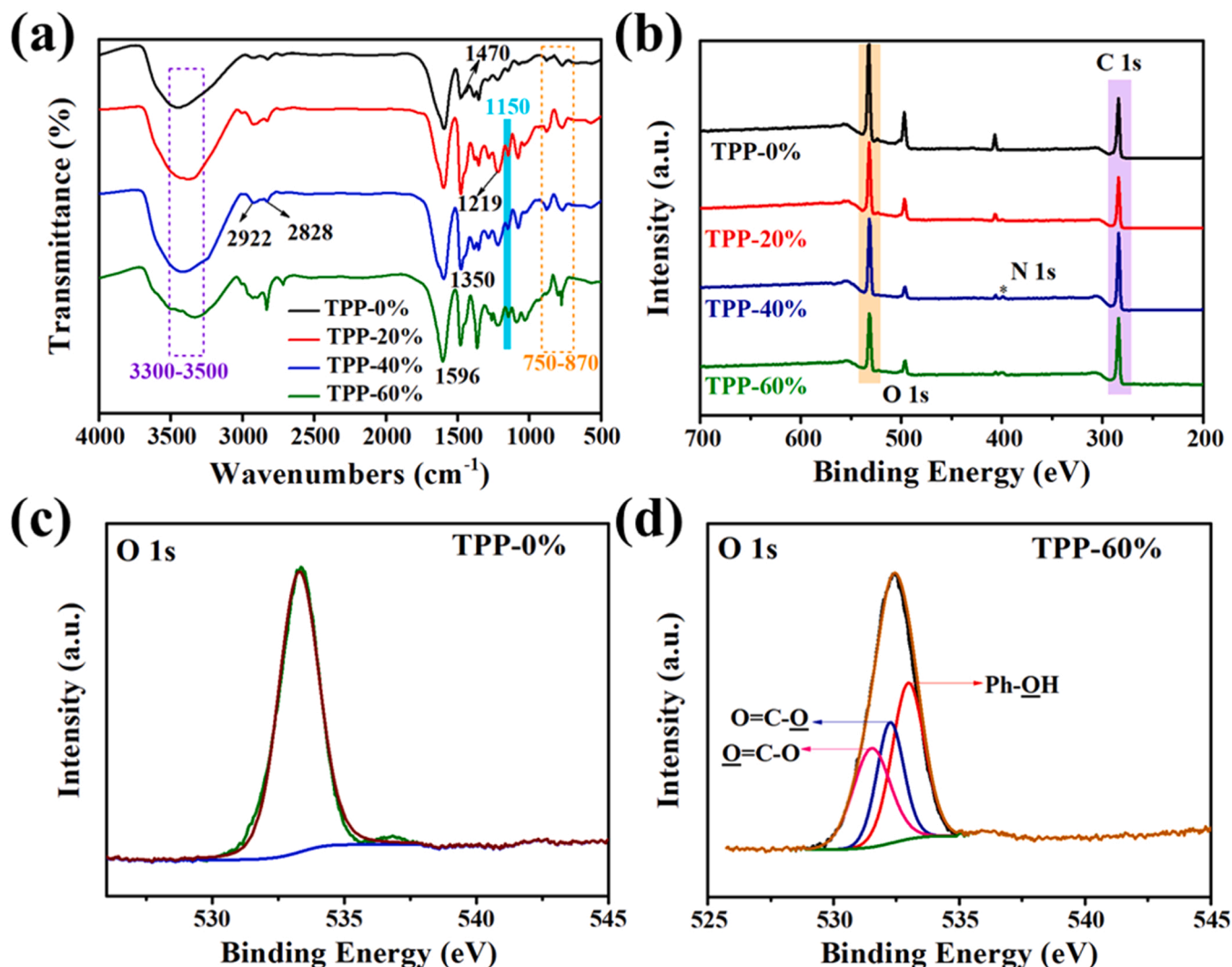


Fig. 1. FT-IR and XPS spectra of the TPP resins, (a): FT-IR of the TPP resins; (b): XPS survey spectra for the surface of TPP resins; (c)-(d): XPS analysis of O 1s of TPP-0 % and TPP-60 %.

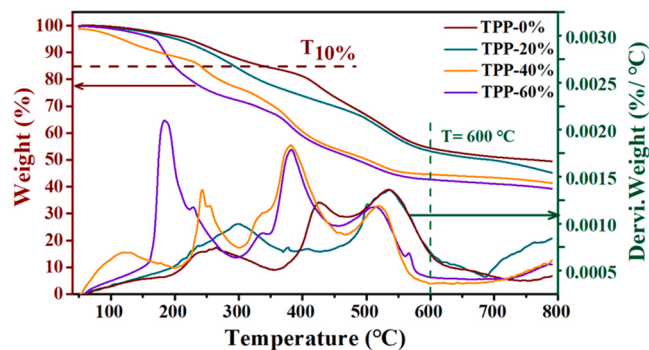


Fig. 2. TG and DTG curves of the cured TPP resins. The left axis shows the percent of the weight (%), and the right axis shows the percent of Deriv. Weight (%/°C).

evaporation of water formed by the condensation reaction of methylol groups; mass loss caused by the condensation reaction between methyl and phenolic hydroxyl and between two hydroxyl functional groups; and mass loss caused by the degradation of methylene [19,23,31]. The TPP resins have similar stages of thermal decomposition. Compared with the TPP-0 % resin, T_{10} % decreased from 275 °C to 256 °C for

TPP-20 % resin, and from 275 °C to 185 °C for the TPP-40 % and TPP-60 %. These results revealed that as the tyrosine level rises, the thermal stability decreases. Furthermore, the carbon residual weight rate at $T = 600$ °C decreased by 55 %, 53 %, 45 %, and 43 %, respectively, as the tyrosine content increased.

Carbon residue rate is a crucial parameter to characterize the thermal stability of materials. The carbon residue of TPP resins decreased as the tyrosine content increased, primarily because tyrosine has only two ortho sites, which reduced the crosslinking density of resin compared with phenol, resulting in the decrease in thermal stability. For TPP-60 %, the decreasing proportion of carbon residue was low and maintained within 10 % compared with TPP-0 %. Table S1 shows that the crosslinking degree of the resins decreases with the increase in tyrosine content. The crosslinking degree of TPP-60 % is lower than that of TPP-0 %, and the thermal stability of TPP-60 % materials with low crosslinking density will be relatively poor because oligomer substances are more accessible to thermal decomposition, resulting in the lowest carbon residue, that is, the thermal property of resins will decrease as the tyrosine content is increased. The loss of thermal stability may be due to the higher cross-linked resin network of TPP-0 % resin, which offers better thermal stability [19,32]. However, the overall thermal stability of all resins is almost similar. The performance of TPP resins suggested that tyrosine can substitute or supplement petroleum-based hazardous phenol in the synthesis of phenolic resins.

3.3. Surface topography

SEM and AFM were employed to examine the surface of TPP resins to see if suitable surface morphology can exhibit an antifouling effect on tiny organisms [4,33]. Fig. 3 shows that several micro-ripples were observed on the surface of the resins. The surface morphology becomes denser and more regular as the tyrosine content increases. The micro-topography images of TPP-0 % in Fig. 3(a) and (a') indicate that some raised ridging structures were irregular and sparse. For TPP-20 % ((b) and (b')) and TPP-40 % ((c) and (c')), the micro-ripples structures formed on the surface are narrow and discontinuous. However, the dense and more regular micro-ripples structures like shells can be observed on the surface of the TPP-60 % ((d) and (d')) with widths ranging from 1 to 2 μm . Fig. 3(a')–(d') shows the roughness (R_a) of the surface resins aside from the observation of the aforementioned surface structure. The R_a of TPP resins decreased as the tyrosine content in the materials increased. The surface roughness of TPP-0 % is 26.5 nm, while the roughness of TPP-60 % is reduced to 8.76 nm. In other words, the surface of TPP-60 % was smoother than that of the other TPP resins; this characteristic offers fewer biological attachment sites and has the best antifouling effect [34,35].

The micro-ripples structures that appeared on the resin surface were examined from two aspects. On one hand, the TPP resins exhibit curing shrinkage because high-temperature curing of phenolic resin involves a polycondensation reaction, releasing part of the small molecule products [36,37], which may cause the surface to shrink. Fig. 3 shows that the surface of TPP-0 % without any tyrosine also has a small amount of micro-ripples structures. On the other hand, TPP resin comprises many hydrophilic tyrosine units with ionic amino and carboxyl residues in the hydrophobic polymer network of phenolic resin. The great polarity difference may result in the microphase separation during the curing process [38–40]. In the curing process, a large number of amino acid residues between the amino and carboxyl groups can form intermolecular forces, such as hydrogen bonding or ionic interaction, forming directional accumulation and creating micro-ripples structures on the surface of the TPP resins. In other words, the hydrophilic domains are assumed to be micro-ripples structure protrusions on the resin surface. The addition of tyrosine to TPP resins enhances the formation of ordered and dense ripples. These two aspects of reasons caused the micro-ripples structures of the resin surface to appear, which provided a basis for the subsequent antifouling properties of materials.

3.4. Surface wettability property

We suspected that the addition of tyrosine affected the hydrophilicity of the TPP resin in addition to the surface topography of materials. Fig. 4 demonstrates that the subsequent experimental results confirmed our hypothesis. The surface energy and contact angles (water, diiodomethane) of the TPP resins are shown in Fig. 4(a). The water contact angle (WCA) of neat phenolic resin was around 84.2° , and after tyrosine was added, the WCA decreased to 82.2° , 73.3° , 55.7° for TPP-20 %, TPP-40 %, TPP-60 %, respectively. Simultaneously, it was evident that the diiodomethane contact angle of the TPP resin increased with the increase in tyrosine content. Fig. 4 shows that with the increase of tyrosine, the number of polar functional groups increases, surface energy increases, and surface hydrophilicity enhances [5,41–44], inducing a strong surface hydration layer that forms a physical barrier to prevent microorganism adhesion on the surfaces [27,45–49]. The TPP-60 % exhibited stronger hydrophilicity and higher surface resistance to microorganisms than TPP-0 % (as shown in Figs. 5–7) [50].

We measured the change of WCA of TPP resins after soaking them in water for 24 h to show that tyrosine hydration in water exists on the surface of the resin. Fig. 4(b) shows how the hydrophilicity improved remarkably as the contact angle declined. Notably, the WCA of other resins decreased from 82.2° , 73.3° , and 55.7° – 66° , 50.7° , and 30.2° for TPP-20 %, TPP-40 %, and TPP-60 %, respectively, after being immersed in water for 24 h, suggesting that the TPP resins exhibited the apparent hydration behavior in water [33,51]. Hydrophilic polymer materials could strongly interact with the surrounding water molecules, generating a dense hydration layer through electrostatic interactions or hydrogen bonds [52]. The presence of a hydration layer could effectively prevent the nonspecific adhesion of biofouling.

3.5. Antifouling performances

3.5.1. Bacteria adhesion

The TPP resins were immersed in *E. coli* and *S. aureus* bacteria at 37°C for 24 h in a fluid environment to investigate the antifouling performance. The anti-adhesion effect was quantified by counting the number of bacterial colonies on agar plates using the spread plate method and estimating the anti-adhesion rate of TPP resins relative to TPP-0 % by Formula (S2). Fig. 5(a) and (b) demonstrate that with the increase of tyrosine content, the number of bacteria on the agar plate decreased and antibacterial adhesion rate improved, indicating that the

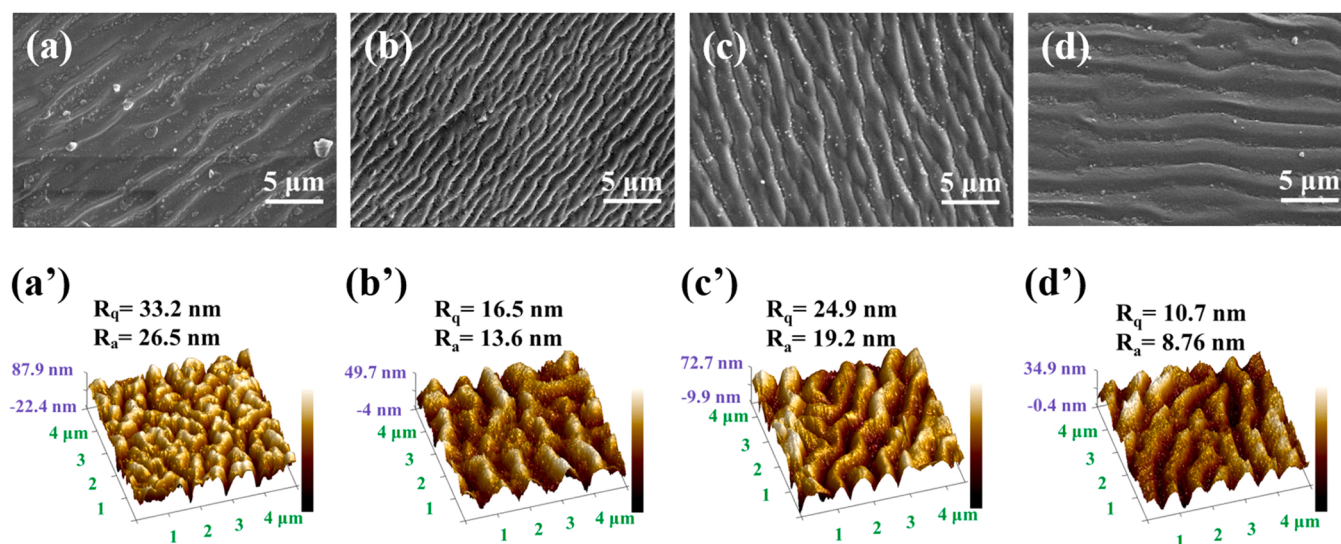


Fig. 3. SEM ((a)–(d)) and AFM 3D images ((a')–(d')) of the cured TPP resins, (a) and (a'): TPP-0 %; (b) and (b'): TPP-20 %; (c) and (c'): TPP-40 %; (d) and (d'): TPP-60 %.

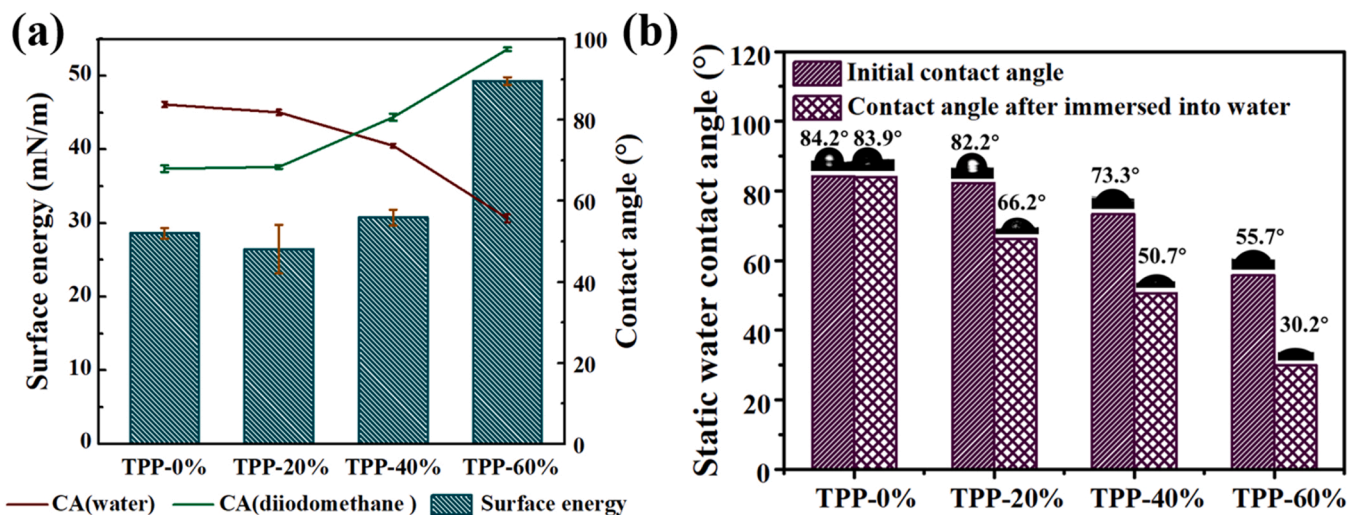


Fig. 4. The change of surface energy, WCA and DCA of the TPP resins with tyrosine content increased (a); the change of the WCA between the initial CA and the CA after soaking in deionized water for 24 h (b).

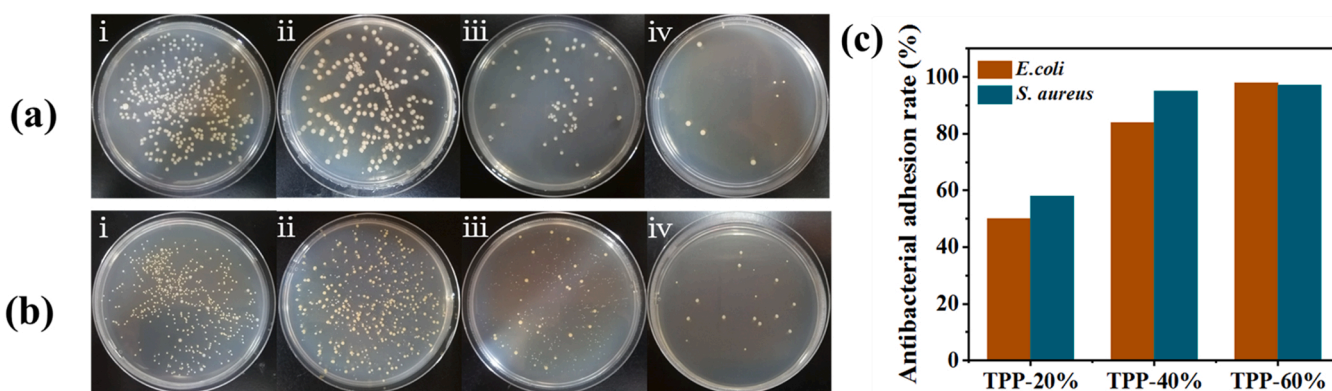


Fig. 5. Agar plate images of the diluted bacterial colonies produced by *E. coli* (a) and *S. aureus* (b) after the detachment from resin surfaces, i-iv were TPP-0 %, TPP-20 %, TPP-40 % and TPP-60 %, respectively; (c): the antibacterial adhesion rate (BAR) to *E. coli* and *S. aureus* of TPP resins, the BAR of TPP-0 % is 0 %.

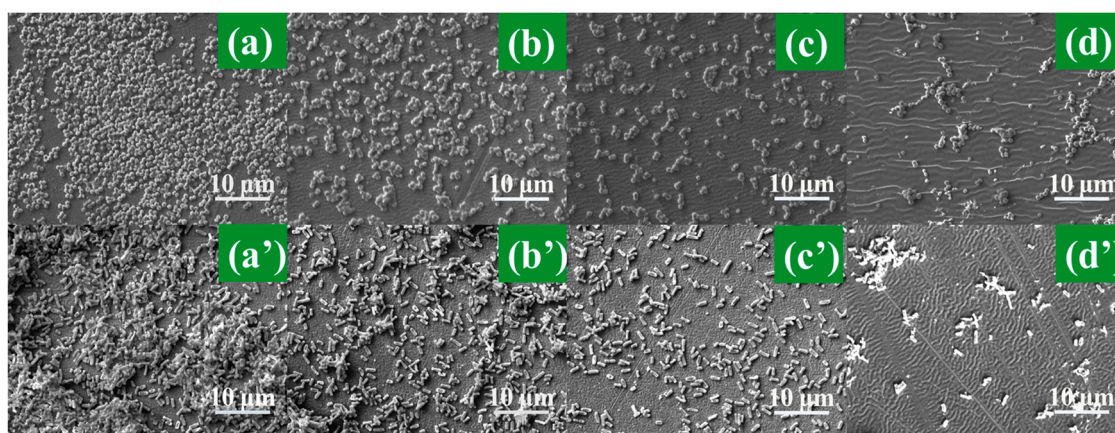


Fig. 6. SEM of bacterial biofilm formation after immersed in the bacteria solution for 7 days; Biofilms formed in *S. aureus* (a-d) and in *E. coli* (a'-d') on TPP-0 %, TPP-20 %, TPP-40 % and TPP-60 %, respectively.

antifouling effect of the TPP resins was enhanced. A large number of *E. coli* and *S. aureus* colonies can be discovered on the agar plates of TPP-0 %, while TPP-60 % contains the least number of bacterial colonies (less than 20). In addition, the antibacterial adhesion rate of TPP-20 %, TPP-40 %, and TPP-60 % toward *E. coli* (*S. aureus*) were 50 % (58 %), 84 %

(95 %), and 98 % (97 %), respectively (as shown in Fig. 5(c)). Thus, TPP-60 % can effectively hinder the adhesion of *E. coli* and *S. aureus*. Furthermore, the number of adhesions of *E. coli* was less than that of *S. aureus*. This may be attributed to the size of the bacteria *E. coli* and *S. aureus*. *S. aureus* is approximately 500 nm in diameter (smaller than

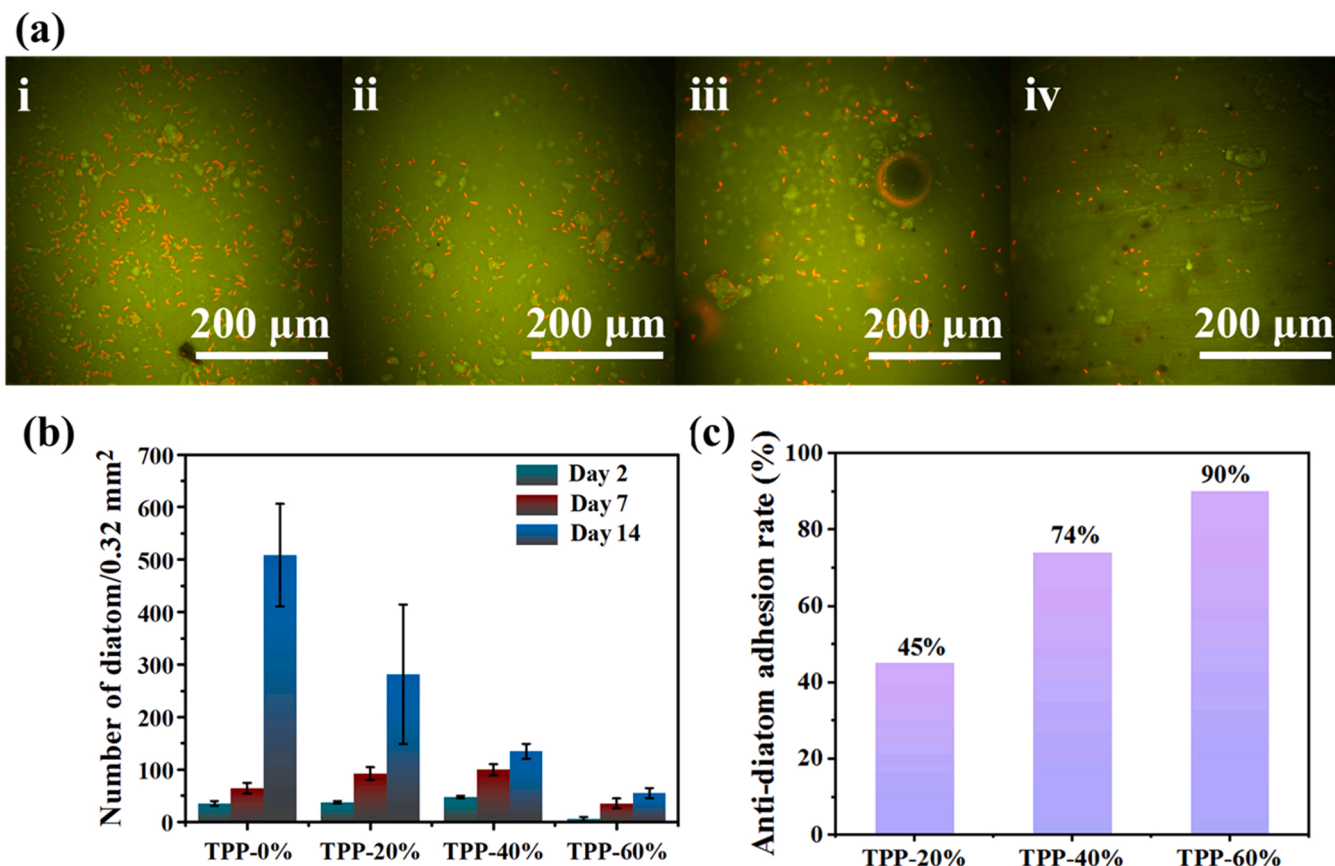


Fig. 7. Fluorescent images of *N. closterium* settled on the TPP resins for 14 days (a), i-iv were TPP-0 %, TPP-20 %, TPP-40 % and TPP-60 %, respectively. The red dots are *N. closterium*. Settlement density data of *N. closterium* on the TPP resins surface (b), density is calculated by counting the algae, number of diatom/0.32 mm². The anti-diatom adhesion rate (DAR) to *N. closterium* of TPP resins (c), the DAR of TPP-0 % is 0 %.

E. coli). However, the size of *E. coli* was similar to the width of micro-ripples structures on the surface of the resin. In terms of the size of bacteria, the morphology of the micro-ripples structures is more convenient for *S. aureus* adhesion. Meanwhile, Fig. S4 shows SEM images of *S. aureus* adherence to the surface of the TPP resins for 24 h, demonstrating its adhesion behavior. The adhesion result of *S. aureus* was consistent with the result of the spread plate method. *S. aureus* adhered to the micro-ripples structure as an attachment site on the surface of TPP-60 %, and the bacteria came together in at least two groups.

To investigate the biofilm formation of bacteria on the resin, SEM was employed to examine the surface after soaking in a bacterial suspension for 7 days. Fig. 6 demonstrates that the coverage rate of biofilm formation by bacterial adhesion decreased with the increase of tyrosine. Moreover, several bacteria adhered to the surface of TPP-0 % and formed adhesions (in Fig. 6(a) and (a')), whereas very few bacteria adhered to the TPP-60 % (in Fig. 6(d) and (d')). This phenomenon may be related to surface morphology and chemical properties of resin surface, which will be explained in Section 3.6 on the possible antifouling mechanism of TPP resins.

3.5.2. Diatom adhesion

A fluorescent microscope was used to observe *N. closterium* adhering to the resin surface in the static antifouling of diatom experiments; the result is shown in Fig. 7(a)–(d). The number of *N. closterium* adhesion of TPP resins decreased as the tyrosine content increased. Quantitatively, after immersion in the medium for 2 days (shown in Fig. S7 (a)), the amount of diatom adhesion of TPP-20 %, TPP-40 %, and TPP-60 % was 35, 37, and 6, respectively. TPP resin began exhibiting an obvious difference in the inhibitory effect on *N. closterium* after 7 days of immersion

(shown in Fig. S7 (b)). After being immersed in the medium for 14 days, the TPP-60 % resin surface adhered to the least diatoms, approximately 50 diatoms/0.32 mm² (shown in Fig. 7(a) and (b)). Fig. 7(c) shows that the diatom anti-adhesion efficiency of the resin improved from TPP-20 % to TPP-60 %. Diatom adhesion results are consistent with bacterial adherence, as expected (Figs. 5 and 6). In short, TPP-60 % provides the best antifouling performance for bacteria and diatoms.

3.6. Possible antifouling mechanism

The possible antifouling mechanism was primarily investigated from two aspects of TPP resin's surface wettability properties and morphology. Previous studies have shown that microorganisms like diatom are more accessible to release from a hydrophilic surface than from a hydrophobic one [35,48]. According to the experimental results, the increased tyrosine content improves hydrophilicity, which results in better antifouling performance (shown in Figs. 5–7). The TPP resins exhibit vital surface hydration resulting in increased hydrophilicity in an aqueous environment (Fig. 4(b)), which is the crucial reason for the nonspecific adhesion of microorganisms. When microorganisms approach the surface of the resin, the hydration layer improved by electrostatic interactions causes a repulsive force, thus reducing the chance for microorganisms to stick to the surface [45,46,51,53,54].

In addition, micro-sized ripples with widths ranging from 1 to 2 μm were observed on the surface of the TPP resins (Fig. 3) that are more similar to that of natural shell microtopography [55,56]. According to the attachment point theory, surface microstructure and roughness can influence microbial attachment. It is tough for microorganisms to settle on their surface when the microstructure is smaller than that of the organism [57]. Furthermore, microorganisms adhered to raised

microstructure are more susceptible to influence by hydrodynamic forces and then harder to stably settle on the surface [58,59]. Thus, the micro-ripples of resin's surface make it challenging for the microorganisms to find appropriate attachment points. Furthermore, when the rough surface of the material comes in contact with microorganisms, the increase of attachment sites on the surface increases the surface adhesion, facilitating microorganism settlement and simple formation of biological contamination [11]. For bacteria, nanoscale roughness is more likely to interact with the structural flagella on their surface [34]. Overall, the TPP-60 % surface possesses random micro-ripples structures and minimum roughness, as seen in Fig. 3 SEM and AFM. These features making TPP-60 % exhibit the best anti-adhesion effect.

Based on the above analysis, the surface hydrophilicity and micro-ripples morphology afford the efficient antifouling performance of the TPP resins.

3.7. Conclusion

In summary, tyrosine was employed as the natural phenol source to synthesize a type of TPP resin with excellent antifouling performance. The WCA decreased to 82.2°, 73.3°, and 55.7° for TPP-20 %, TPP-40 %, and TPP-60 %, respectively, with the addition of tyrosine. After soaking in water for 24 h, the static WCA of the TPP resins continued to decrease because of the hydrophilic surface's hydration layer. Furthermore, the surface morphology becomes denser and more regular as the tyrosine content increases, and the surface roughness (Ra) decreased. The surface of TPP-60 % resin forms dense and more regular micro-ripples structures with a width of 1–2 µm, and the Ra decreases to 8.76 nm, which provides fewer adhesion points for microorganisms. Based on the micro-phase separation of tyrosine and phenolic resin and the polycondensation of phenolic resin, the TPP resins exhibit the micro-ripples microtopography like the shell. The antifouling experimental results showed that TPP-60 % exhibited the best antifouling effect, and the anti-adhesion efficiency to bacteria and diatoms reached 95 % for 24 h and 90 % after 14 days. The surface hydrophilicity and micro-ripples morphology make the TPP-60 % excellent antifouling performance compared with traditional phenolic resin. The TPP resins also have a practical value to broaden the applications of phenolic resins and offer crucial guidance for developing the antifouling field in the future.

CRediT authorship contribution statement

Xuan Yuan: Performing the experiments, Visualization, Data collection and analysis, Writing – original draft. **Yudan Wang:** Investigation, Data discussion, Resources. **Lijia Liu:** Supervision, Funding acquisition, Data discussion, Writing – review & editing. **Hongxing Dong:** Data discussion, Writing – review & editing. **Guoxing Yang:** Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfb.2022.112672.

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